

Lab 3 DFIR : Win File sys and registry

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-----: Questions :-----

Part I :

Explain the windows file system.

What kind of commands are used in CMD for DFIR ?

Perform - Use FTK / Autopsy to collect data from files

Part II :

Explain the windows Registry and hives.

Which are the main registry hives and what are their uses?

Perform – Demonstrate how to access the main hives?

----- Part I -----

1. Explain the windows file system.

The NTFS (New Technology File System) is designed to provide robust security, reliability, and scalability, making it suitable for modern computing needs. It incorporates several advanced features that are invaluable during Digital Forensics and Incident Response:

- **File Permissions:** NTFS uses Access Control Lists (ACLs) to define granular permissions for files and folders. These permissions help investigators determine who had access to specific files.
- **File Compression:** By compressing data, NTFS reduces storage requirements, which can influence disk usage patterns relevant to forensics.
- **Encryption:** EFS (Encrypting File System) is a built-in NTFS feature allowing file encryption, often used to protect sensitive data. Forensic investigators need to deal with encrypted files during their analysis.
- **Journaling:** NTFS maintains a transactional log, tracking metadata changes. This can provide clues about changes made to files or directories before a system crash or power failure.

2. What kind of commands are used in CMD for DFIR ?

When investigating incidents, understanding and navigating the filesystem effectively is crucial. Below are commonly used commands with additional insights:

- **dir**: Lists directory contents, including hidden/system files with `dir /a`.
 - Example: `dir C:\Windows\System32 /a` lists critical system files.
 - For forensic triage, combine `dir` with date filters to locate files modified within a suspicious timeframe
- **fsutil**: Offers deep insights into NTFS-specific details.
 - Example: `fsutil fsinfo ntfsinfo C:` provides NTFS version and features used on the volume.
 - Investigators may also use `fsutil volume querycluster` to find files linked to specific disk clusters, helping in file carving and data recovery.
- **chkdsk**: Identifies file system corruption or anomalies, potentially revealing signs of sabotage or disk tampering.
- **xcopy**: Allows recursive copying with preservation of attributes. The `/O` option includes file ownership and ACL information.
- **attrib**: Critical for discovering hidden or tampered files.
 - Example: Files marked with the "hidden" or "system" attribute may indicate malware attempts to evade detection.
- **icacls**: Helps analyze file access rights, which can reveal unauthorized modifications.
- **PowerShell's Get-Item**: Provides detailed file metadata, useful in creating a timeline of events.
 - Example: `Get-Item C:\path\file.txt | Select-Object Name, Length, Attributes` shows file size and attributes.

Autopsy and Filesystem Analysis

Autopsy simplifies filesystem analysis for forensic professionals. Beyond basic functions, it provides:

- **Data Visualization**: Interactive timelines allow investigators to correlate file access and modification with incident events.
- **File Hashing**: Generates MD5/SHA-1 hashes to verify file integrity or compare with known malware databases.
- **Registry Hive Extraction**: Assists in analyzing Windows registry hives stored on disk.

```
Select Administrator: Command Prompt
dir C:\Users\
Volume in drive C is OS
Volume Serial Number is E064-9E68

Directory of C:\Users

12/22/2024  05:50 AM  <DIR>      .
01/18/2025  01:13 PM  <DIR>      Admin
09/14/2024  10:43 PM  <DIR>      Public
               0 File(s)      0 bytes
               3 Dir(s)  1,013,047,328 bytes  1 file

fsutil fsinfo volumeinfo C:
Volume Name : OS
Volume Serial Number : 0xe0649e68
Max Component Length : 255
File System Name : NTFS
Is ReadWrite
Not Thinly-Provisioned
Supports Case-sensitive filenames
Preserves Case of filenames
Supports Unicode in filenames
Preserves & Enforces ACL's
Supports file-based Compression
Supports Disk Quotas
Supports Sparse files
Supports Reparse Points
Returns Handle Close Result Information
Supports POSIX-style Unlink and Rename
Supports Bypass IO
Supports Case-Sensitive Directories
Supports Object Identifiers
Supports Encrypted File System
Supports Named Streams
Supports Transactions
Supports Hard Links
Supports Extended Attributes
Supports Open By FileID
Supports USN Journal

Administrator: Command Prompt
attrib xwreg.dll
A                C:\Windows\System32\xwreg.dll

chkdsk C:
The type of the file system is NTFS.
Volume label is OS.

WARNING! /F parameter not specified.
Running CHKDSK in read-only mode.

Stage 1: Examining basic file system structure ...
1010432 file records processed.
File verification completed.
Phase duration (File record verification): 11.89 seconds.
21215 large file records processed.
Phase duration (Orphan file record recovery): 8.34 milliseconds.
0 bad file records processed.
Phase duration (Bad file record checking): 2.16 milliseconds.

Stage 2: Examining file name linkage ...
582 reparse records processed.

1332136 index entries processed.

Index verification completed.
Phase duration (Index verification): 20.01 seconds.

0 unindexed files scanned.
Phase duration (Orphan reconnection): 1.69 seconds.

0 unindexed files recovered to lost and found.
Phase duration (Orphan recovery to lost and found): 0.89 milliseconds.

582 reparse records processed.
Phase duration (Reparse point and Object ID verification): 5.17 milliseconds.

Stage 3: Examining security descriptors ...
Security descriptor verification completed.
Phase duration (Security descriptor verification): 32.30 milliseconds.
```

3. Perform - Use FTK / Autopsy to collect data from files

The following screenshots depict the loading of a drive called F onto autopsy for analysis it is found to have :

- 2 Deleted files
 - An Image file
 - A Word document

Upon further inspection we find that the document contains text depicting a cuda based python GUI script for **bruteforcing a protected document**.

Further analysis helps us to find that the same document has another application file called **Attack.exe** embedded in the document which probably is [actually is] the implementation of the python script

There is another image document attached of a building with corresponding location data which might give us insights into the where about of the creation of the documents

Partition F - Autopsy 4.21.0

Case View Tools Window Help

Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Directory Tree

Listing

/img_F:

21 Results

Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size
System Volume Information				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	5
\$AttrDef				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2
\$BadClus				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	0
\$BadClus:\$Bad				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	5
\$Bitmap				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	1
\$Boot				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	8
\$LogFile				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	1
\$MFT				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2
\$MFTMirr				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	4
\$Secure\$SDS				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2
\$UpCase				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	1
\$UpCase\$Info				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	3
\$Volume				2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	2025-01-22 12:40:15 IST	0
DECR - Copy.docx				2025-01-22 12:42:44 IST	2025-01-22 12:42:44 IST	2025-01-22 12:42:30 IST	2025-01-22 12:42:30 IST	9
IMG_20220925_065550.jpg				2025-01-22 12:42:44 IST	2025-01-22 12:42:44 IST	2025-01-22 12:42:40 IST	2025-01-22 12:42:40 IST	2

Data Content

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Partition F - Autopsy 4.21.0

Case View Tools Window Help

Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Directory Tree

Listing

File System

2 Results

Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size
DECR - Copy.docx				2025-01-22 12:42:44 IST	2025-01-22 12:42:44 IST	2025-01-22 12:42:30 IST	2025-01-22 12:42:30 IST	92526
IMG_20220925_065550.jpg				2025-01-22 12:42:44 IST	2025-01-22 12:42:44 IST	2025-01-22 12:42:40 IST	2025-01-22 12:42:40 IST	25098

Data Content

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Partition F - Autopsy 4.21.0

Case View Tools Window Help

Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search

Directory Tree

- Data Sources
 - F:1 Host
 - F:
 - \$OrphanFiles (0)
 - \$CarvedFiles (1)
 - \$Extend (7)
 - \$RECYCLE.BIN (3)
 - \$Unalloc (6)
 - System Volume Information (24)
- File Views
 - File Types
 - By Extension
 - By MIME Type
 - Deleted Files
 - File System (2)
 - All (3)
 - MB File Size
 - Data Artifacts
 - Analysis Results
 - EXIF Metadata (1)
 - Keyword Hits (20)
 - User Content Suspected (1)
 - OS Accounts
 - Tags
 - Score
 - Reports

Listing

File System

Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size
DECR - Copy.docx				2025-01-22 12:42:44 IST	2025-01-22 12:42:44 IST	2025-01-22 12:42:30 IST	2025-01-22 12:42:30 IST	92526
IMG_20220925_065550.jpg				2025-01-22 12:42:44 IST	2025-01-22 12:42:44 IST	2025-01-22 12:42:40 IST	2025-01-22 12:42:40 IST	25098

Data Content

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of - Page Matches on page: - of - Match 100% Reset Text Source: File Text

System-Infiltration : 2 : Brute Force

The following documents contains the specific details of the method of brute forcing a protected document.

The specifications of the document are those of the

- Creation of the target document produced
- Production of a python script to analyse and brute force the attack surface
- Execution of the python GUI application and entering relevant details
- Monitoring hardware use and wait for completion of the programs

Target Document

The document displays a box to enter the password that we do not know as of now.

The document was created at 9/12/24, 10:20:55 PM by CSDL at NSE

Technical specifications of the password : PAN_NO.upper()

Python Script

- The python script is a CUDA enabled acceleration cracking program aimed at enhancing brute force speeds.

Partition F - Autopsy 4.21.0

Case View Tools Window Help

Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search

Directory Tree

- Data Sources
 - F:1 Host
 - F:
 - \$OrphanFiles (0)
 - \$CarvedFiles (1)
 - \$Extend (7)
 - \$RECYCLE.BIN (3)
 - \$Unalloc (6)
 - System Volume Information (24)
- File Views
 - File Types
 - By Extension
 - By MIME Type
 - Deleted Files
 - File System (2)
 - All (3)
 - MB File Size
 - Data Artifacts
 - Analysis Results
 - EXIF Metadata (1)
 - Keyword Hits (20)
 - User Content Suspected (1)
 - OS Accounts
 - Tags
 - Score
 - Reports

Listing

File System

Table Thumbnail Summary

Save Table as CSV

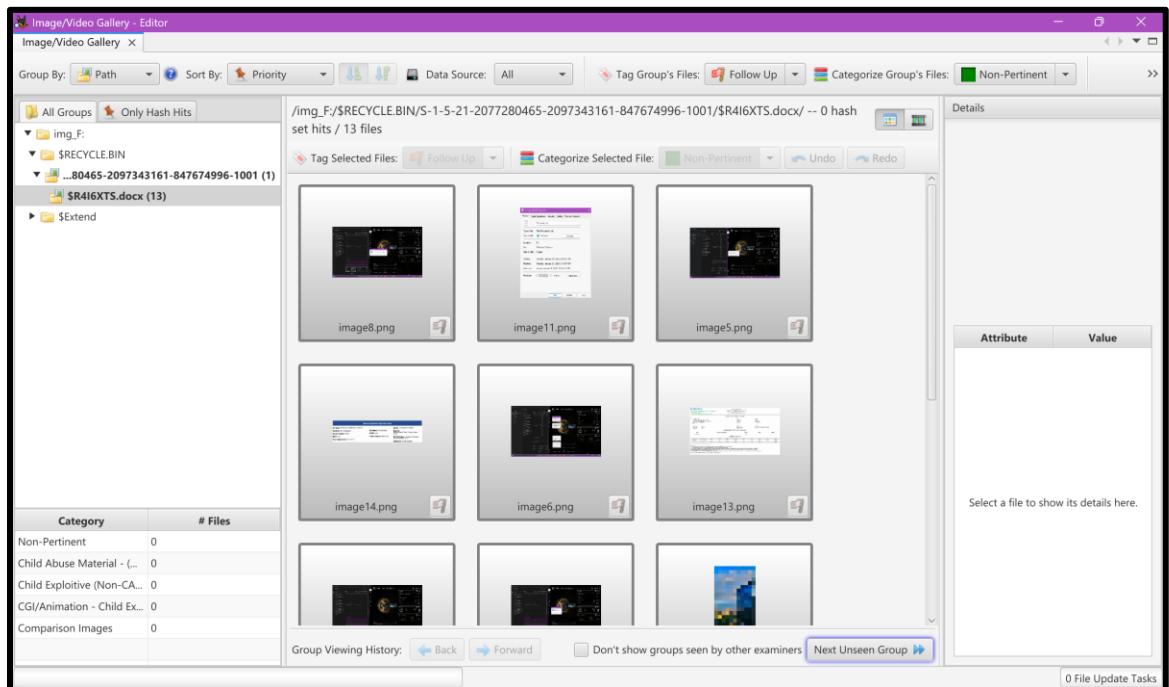
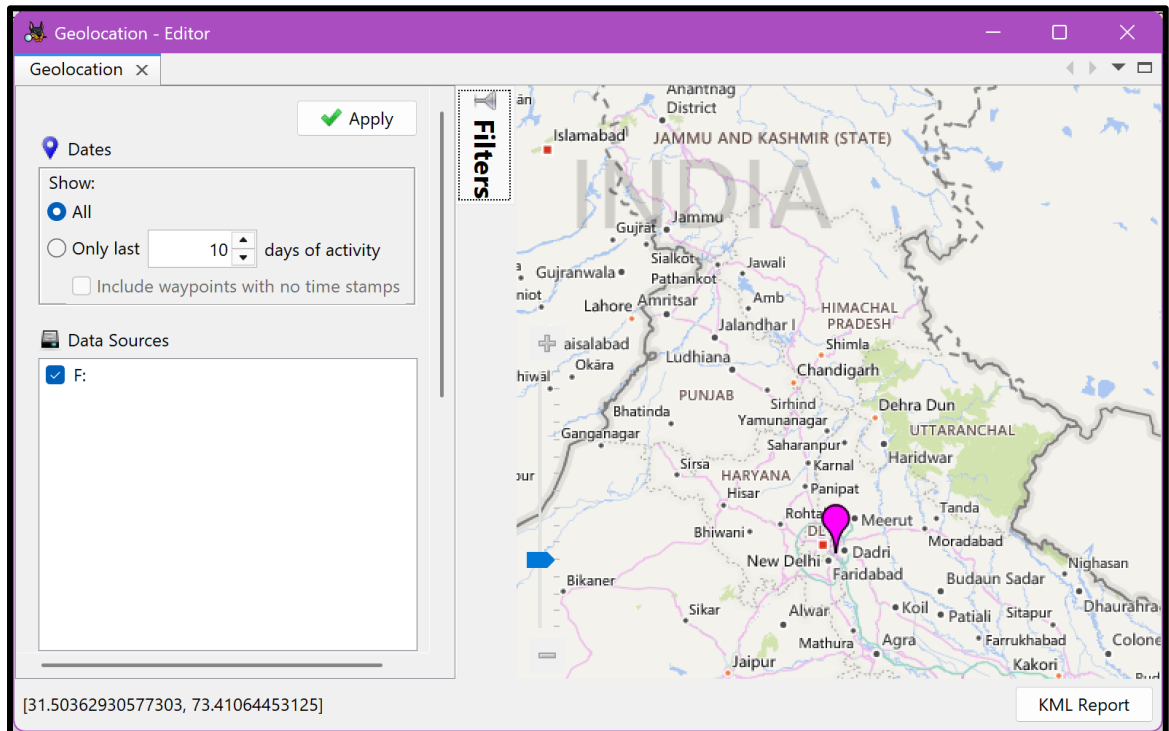
Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size
DECR - Copy.docx				2025-01-22 12:42:44 IST	2025-01-22 12:42:44 IST	2025-01-22 12:42:30 IST	2025-01-22 12:42:30 IST	92526
IMG_20220925_065550.jpg				2025-01-22 12:42:44 IST	2025-01-22 12:42:44 IST	2025-01-22 12:42:40 IST	2025-01-22 12:42:40 IST	25098

Data Content

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

0% 8% Reset Tags Menu





Partition F - Autopsy 4.21.0

Case View Tools Window Help

Directory Tree

- Data Sources
 - F: 1 Host
 - \$OrphanFiles (0)
 - \$CarvedFiles (1)
 - \$Extend (7)
 - \$RECYCLE.BIN (3)
 - \$Unalloc (6)
 - System Volume Information (24)
- File Views
 - File Types
 - By Extension
 - By MIME Type
 - Deleted Files
 - File System (2)
 - All (3)
 - MB File Size
 - Data Artifacts
 - Analysis Results
 - EXIF Metadata (1)
 - Keyword Hits (20)
 - User Content Suspected (1)
 - OS Accounts
 - Tags
 - Score
 - Bad Items (0)
 - Suspicious Items (9)
 - Reports

Suspicious Items

Table Thumbnail Summary

Save Table as CSV

Source	Type	Path	Created Date
Unalloc_4_3897573376_4971315200	File	/img_F/\$Unalloc/Unalloc_4_3897573376_4971315200	
\$R416XTS.docx	File	/img_F/\$RECYCLE.BIN/S-1-5-21-2077280465-2097...	2025-01-22 12:42:44 IST
image13.png	File	/img_F/\$RECYCLE.BIN/S-1-5-21-2077280465-2097...	
Attack.exe	File	/img_F/\$RECYCLE.BIN/S-1-5-21-2077280465-2097...	
f0892182.swf	File	/img_F/\$CarvedFiles/1/f0892182.swf	

Data Content

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Case	Data Source Name	File Name
MaliciousPDF	LogicalFileSet2	attack.exe
	LogicalFileSet3	

Common Properties

Type: File MD5
Value: 72a378783e7ddfc2772a56d0c7f5900
Known Status: unknown

File Details

File Path: /decr - copy.docx/attack.exe

Data Source Details

Name: LogicalFileSet3

Case Details

Name: MaliciousPDF
Created Date: 2025/01/21 14:36:04 (IST)

Central Repository Starting Date: 2025/01/11 13:02:05 (IST)

Found 2 instances in 1 cases and 2 data source

Partition F - Autopsy 4.21.0

Case View Tools Window Help

Directory Tree

- Data Sources
 - F: 1 Host
 - \$OrphanFiles (0)
 - \$CarvedFiles (1)
 - \$Extend (7)
 - \$RECYCLE.BIN (3)
 - \$Unalloc (6)
 - System Volume Information (24)
- File Views
 - File Types
 - By Extension
 - By MIME Type
 - Deleted Files
 - File System (2)
 - All (3)
 - MB File Size
 - Data Artifacts
 - Analysis Results
 - EXIF Metadata (1)
 - Keyword Hits (20)
 - User Content Suspected (1)
 - OS Accounts
 - Tags
 - Score
 - Bad Items (0)
 - Suspicious Items (9)
 - Reports

Suspicious Items

Table Thumbnail Summary

Save Table as CSV

Source	Type	Path	Created Date
Unalloc_4_3897573376_4971315200	File	/img_F/\$Unalloc/Unalloc_4_3897573376_4971315200	
\$R416XTS.docx	File	/img_F/\$RECYCLE.BIN/S-1-5-21-2077280465-2097...	2025-01-22 12:42:44 IST
image13.png	File	/img_F/\$RECYCLE.BIN/S-1-5-21-2077280465-2097...	
Attack.exe	File	/img_F/\$RECYCLE.BIN/S-1-5-21-2077280465-2097...	
f0892182.swf	File	/img_F/\$CarvedFiles/1/f0892182.swf	

Data Content

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Page: 1 of 5342 Page Go to Page: 1 Jump to Offset Launch in HxD

0x0000060:	8D 05 7A A4 02 00 48 8D	0D 67 A4 02 00 8B 10 E8	...S...H...g... ..
0x00000700:	0C 16 00 00 E9 30 01 00	00 4C 89 A4 24 D0 00 00	...D...L...S... ..
0x00000710:	00 41 BF FF FF FF FF 44	8B 63 08 0F 1F 44 00 00	...A...D...C...D... ..
0x00000720:	B8 00 20 00 00 49 8B DC	4C 3B E0 4C 8B CF BA 01	...I...L...I... ..
0x00000730:	00 00 00 49 8B CD 48 0F	47 D8 4C 8B C3 E8 CA F0	...I...H...G...L... ..
0x00000740:	00 00 48 3B C3 0F 85 E6	00 00 00 48 8B CF E8 2D	...H... ..
0x00000750:	EE 00 00 85 C0 0F 85 D6	00 00 00 4C 2B E3 89 5C	L... ..
0x00000760:	24 28 4C 89 6C 24 20 66	0F 1F 04 00 00 00 00 00	\$(L18 f... ..
0x00000770:	BB 00 20 00 00 48 89 6C	24 30 33 D2 89 5C 24 38	...R.1903... ..
0x00000780:	48 8D 4C 24 20 E8 A6 8E	00 00 8B F8 41 BF FF FF	W.L8
0x00000790:	FF FF 8D 48 04 83 F9 02	76 7C 83 F8 02 74 72 8B	...H... ..
0x000007a0:	4C 24 38 48 2B D9 4D 85	F6 74 29 4D 8B CE 4C 8B	L8BH...M... ..
0x000007b0:	C3 BA 01 00 00 00 48 8B	CD E8 8E F7 00 00 48 3B	H... ..
0x000007c0:	C3 75 0C 49 8B CE E8 B5	ED 00 00 85 C0 74 1B 41	...U... ..t...A... ..
0x000007d0:	8B FF E8 42 48 85 F6 74	11 4C 8B C3 48 8B D5 48	...BH... ..L... ..H... ..
0x000007e0:	8B CE E8 B9 8A 02 00 48	03 F3 83 7C 24 38 00 0F	H... ..I80... ..
0x000007f0:	84 7B FF FF FF 83 FF 01	74 12 4D 85 E4 74 17 48	t... ..M... ..t... ..H... ..
0x00000800:	8B BC 24 C0 00 00 00 E9	14 FF FF FF 45 33 FF E8	S...
0x00000810:	20 BF FD FF FF FF 48 8B	94 24 C8 00 00 00 48 8D	H... ..S...E3... ..

1. Explain the windows Registry and hives.

The Windows Registry

The Windows Registry is a hierarchical database that stores low-level settings for the Windows operating system, applications, hardware, and user preferences. It is a critical component of the Windows operating system that ensures proper system functioning. The registry organizes its data into keys (similar to folders) and values (similar to files), where the keys act as containers, and the values store configuration information.

The registry is crucial for several reasons:

- It defines how the OS interacts with hardware and software.
- It maintains settings for user profiles and system configurations.
- It serves as a storage location for many system-related events, making it essential for forensic analysis.

Registry Hives

A **registry hive** is a logical grouping of registry keys, subkeys, and values that represent different parts of the registry. Each hive corresponds to a specific area of the system, such as user preferences, hardware configurations, or software settings. Registry hives are stored as physical files on disk, making it possible to analyze them even when a system is offline.

For example:

- SYSTEM hive is stored as a file named SYSTEM under C:\Windows\System32\Config.
- NTUSER.DAT is a hive file for individual user profiles.

2. Which are the main registry hives and what are their uses?

Hive Name	Abbreviation	Description	Uses in DFIR
HKEY_CLASSES_ROOT	HKCR	Stores information about file associations and OLE (Object Linking and Embedding) objects.	Helps identify file types associated with specific programs or malicious file association changes.
HKEY_CURRENT_USER	HKCU	Contains user-specific settings for the currently logged-in user, such as desktop preferences, application configurations, and browsing history.	Reveals user activity, such as recently used files or applications, and custom settings.
HKEY_LOCAL_MACHINE	HKLM	Contains global system settings, including hardware configurations, installed software, and services.	Critical for analyzing system-wide configurations, installed programs, and services, including malicious additions.
HKEY_USERS	HKU	Stores information for all user profiles on the system.	Provides insights into multiple user profiles, including activity and settings for non-active accounts.
HKEY_CURRENT_CONFIG	HKCC	Contains settings related to the hardware profile in use during the current session.	Helps identify the active hardware configuration, useful for troubleshooting device-related issues.
HKEY_PERFORMANCE_DATA	Not abbreviated	Contains real-time performance data for system monitoring (rarely used in DFIR).	Rarely analyzed in DFIR but could assist in performance or resource monitoring during an incident.

3. Perform – Demonstrate how to access the main hives?

Using Registry Editor (GUI)

1. **Open Registry Editor:**
 - Press Win + R, type regedit, and press Enter.
 - Click **Yes** if prompted by User Account Control (UAC).
2. **Main Registry Hives:** In the left-hand pane of Registry Editor, you'll see the following main registry hives:
 - **HKEY_CLASSES_ROOT (HKCR):** Stores information about file types, file associations, and COM objects.
 - **HKEY_CURRENT_USER (HKCU):** Contains settings for the currently logged-in user.
 - **HKEY_LOCAL_MACHINE (HKLM):** Stores settings that apply to the entire computer.
 - **HKEY_USERS (HKU):** Contains user profile information for all user accounts on the system.
 - **HKEY_CURRENT_CONFIG (HKCC):** Contains information about the current hardware configuration.
3. **Expand Hives:**
 - Click the dropdown arrow or double-click a hive to expand it and view its subkeys.

Using Command Prompt

While less common, you can query the registry with the reg command.

List Registry Keys

Example:

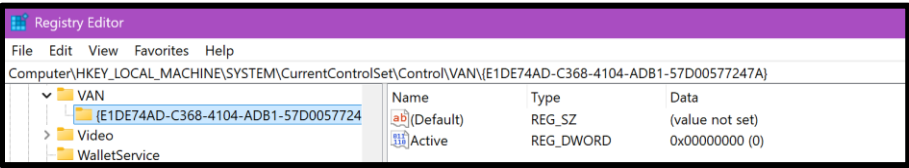
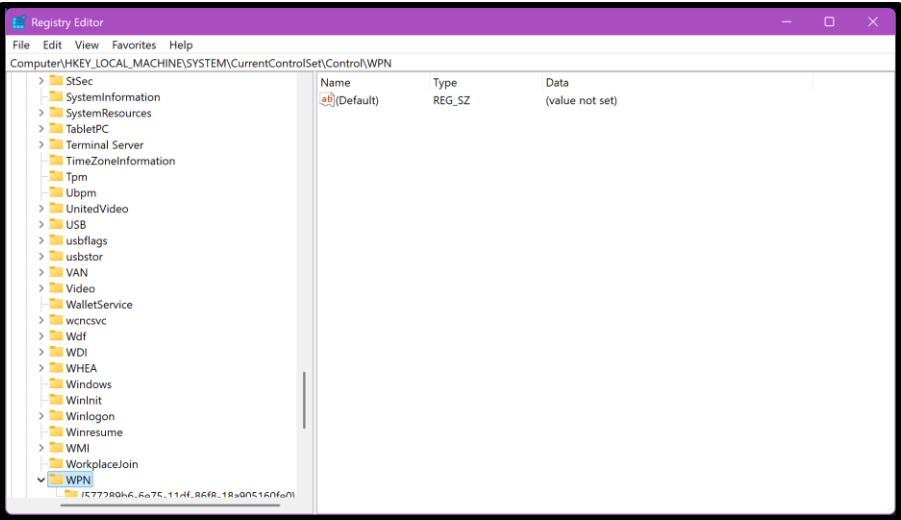
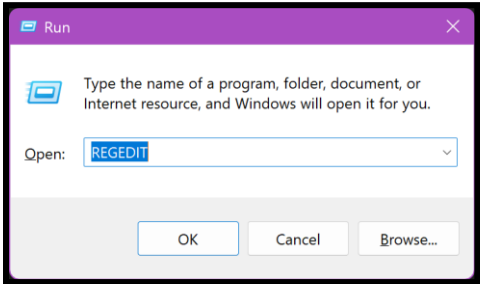
```
reg query HKEY_LOCAL_MACHINE
```

Query a Specific Key

Example:

```
reg query HKEY_LOCAL_MACHINE\SOFTWARE
```

These methods provide access to and allow manipulation of the Windows Registry. Ensure you proceed cautiously, as changes to the registry can impact system stability.



```
C:\Windows\system32\cmd.e  x  +  v  -  □  x
Microsoft Windows [Version 10.0.26100.2894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>reg query HKEY_LOCAL_MACHINE

HKEY_LOCAL_MACHINE
    ServiceLastKnownStatus    REG_DWORD    0x2

HKEY_LOCAL_MACHINE\BCD00000000
HKEY_LOCAL_MACHINE\DRIVERS
HKEY_LOCAL_MACHINE\HARDWARE
HKEY_LOCAL_MACHINE\SAM
HKEY_LOCAL_MACHINE\SECURITY
HKEY_LOCAL_MACHINE\SOFTWARE
HKEY_LOCAL_MACHINE\SYSTEM
HKEY_LOCAL_MACHINE\WindowsAppLockerCache

C:\Users\Admin>reg query HKEY_LOCAL_MACHINE\SOFTWARE

HKEY_LOCAL_MACHINE\SOFTWARE\AGEIA Technologies
HKEY_LOCAL_MACHINE\SOFTWARE\AMD
HKEY_LOCAL_MACHINE\SOFTWARE\AMDL0G
HKEY_LOCAL_MACHINE\SOFTWARE\Apple Inc.
HKEY_LOCAL_MACHINE\SOFTWARE\ASIO
HKEY_LOCAL_MACHINE\SOFTWARE\ASUS
HKEY_LOCAL_MACHINE\SOFTWARE\ASUSHitManager
HKEY_LOCAL_MACHINE\SOFTWARE\ATI
HKEY_LOCAL_MACHINE\SOFTWARE\ATI Technologies
HKEY_LOCAL_MACHINE\SOFTWARE\AUUP
HKEY_LOCAL_MACHINE\SOFTWARE\Bytedance
```